

Sudarshan Babu

Research Scientist - Generative Modeling

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Research Summary

Research scientist developing generative models for data-scarce domains. Specializes in 3D generation (scenes, small molecules, cryo-ET volumes) and single-cell modeling using few-shot learning methods, hypernetworks, and meta-learning.

Core Expertise

Research Diffusion models, 3D generation, small molecule design, single cell modelling
Methods Hypernetworks, implicit neural representations (INRs), transformers, meta-learning

Professional Experience

- Sep 2025–Present **Research Scientist**, *Chan Zuckerberg Biohub Chicago*, Chicago, IL.
◦ Developed scLDM, a latent diffusion model for single-cell data generation
◦ Built implicit neural field model for cryo-ET volume generation
- Jan 2025–Sep 2025 **AI Fellow**, *Chan Zuckerberg Biohub Chicago*, Chicago, IL.
◦ Developed HyperDiffusionFields (HyDiF), combining diffusion models with hypernetworks for implicit molecular neural fields generation
- Oct 2021–Jan 2022 **Research Scientist Intern**, *Amazon*, Bellevue, WA..
◦ Designed curriculum learning strategies for efficient neural network training
- Feb 2021–Jun 2021 **Research Scientist Intern**, *NVIDIA*, Remote.
◦ Built systems for training models on non-stationary long-tail distributions
◦ Contributed to 4 patent applications in adaptive learning and distributed training systems

Education

- Oct 2019–2025 **Ph.D., Computer Science**, *Toyota Technological Institute at University of Chicago*, Chicago, IL, GPA: 3.90/4.0.
◦ Advisors: Prof. Michael Maire, Prof. Greg Shakhnarovich
◦ Dissertation: **Acquiring and Adapting Priors for Novel Tasks via Neural Meta-Architectures**

Papers

- NeurIPS WS '25 **HyperDiffusionFields (HyDiF): Diffusion-Guided Hypernetworks for Learning Implicit Molecular Neural Fields**
Sudarshan Babu et al.*
- NeurIPS WS '25 **Scalable Latent Diffusion Model for Single-Cell Gene Expression Data**
Giovanni Palla et al.
- ICML '24 **HyperFields: Towards Zero-Shot Generation of NeRFs from Text**
Sudarshan Babu et al.*
- SIGGRAPH Asia '24 **iSeg: Interactive 3D Segmentation via Interactive Attention**
Itai Lang et al.*
- NeurIPS '21 **Online Meta-Learning via Learning with Layer-Distributed Memory**
Sudarshan Babu et al.
- CVPR '21 **Domain-Independent Dominance of Adaptive Methods**
Pedro Savarese et al.
- Preprint **HyperNetwork Designs for Improved Classification and Robust Meta-Learning**
Sudarshan Babu, Pedro Savarese, Michael Maire

Service & Leadership

- Reviewing CVPR, ICCV, ECCV, NeurIPS, ICLR, SIGGRAPH, RECOMB
- Teaching Teaching Assistant for Machine Learning (Prof. Greg Shakhnarovich), Fall 2020
- Organization Co-organizer: Computational Biology Workshop @ TTIC (Sep 2024), Annual TTI-C Student Workshop (Feb 2020), Computer Vision Reading Group (Oct 2020 – Feb 2024)